Cheduluri Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.67
G.R.C Jr. College, R.C .Puram	Class XII (AP Intermediate board)	2009	96.70%
Dr. D.S. Raju AP Res. School, Bhupathipalem	Class X (AP SSC)	2007	90.83%

POSITIONS OF RESPONSIBILITY

- **RTL & FPGA Team Lead / PhD Research Scholar** - Leading RTL design, FPGA implementation, and hardware-based research activities, including task allocation, technical coordination, and project execution.
- **B.Tech Project Team Lead** - Led and mentored B.Tech student project teams, providing technical guidance in digital design, RTL development, FPGA implementation, verification, and project documentation.
- **Teaching Assistant - Digital Design Course** - Served as a Teaching Assistant for the Digital Design course, supporting laboratory sessions, assisting students with digital logic and RTL design concepts, and providing guidance on design and verification activities.
- Delivered a lecture on the **Cadence Virtuoso simulator** in the SMDP workshop conducted by VNIT.
- Teaching Assistant for Analog and RF IC Design Courses 2017 -2019.

SKILLS

- **RTL Design:** Verilog HDL, SystemVerilog, RTL coding, FSM design, datapath, and control-unit design
- **FPGA:** FPGA architecture, FPGA-based prototyping, synthesis, implementation, timing analysis, hardware debugging
- **Digital Design:** Combinational and sequential logic, pipelining, finite-state machines, memories, interfaces, and digital system architecture
- **Verification:** RTL simulation, functional verification, testbench development, waveform analysis, debugging
- **EDA Tools:** Xilinx Vivado, Intel Quartus, ModelSim/QuestaSim, Vivado Vitis, ISE, DC, TCAD (Sentaurus- Synopsys), Virtuoso, Genus, Innovus(Cadence), MATLAB, Ngspice,
- **Hardware:** FPGA development boards, digital system prototyping, hardware-software interfacing
- **Programming:** C/C++, Python, MATLAB
- **Research:** Hardware acceleration, RTL optimization, low-power digital design, FPGA-based implementation, biomedical/AI hardware systems

Bus Protocols & Interfaces

- **AXI (AMBA):** AXI4, AXI4-Lite, AXI4-Stream
- **AHB (AMBA):** AMBA AHB
- **APB (AMBA):** AMBA APB
- **SPI:** Serial Peripheral Interface
- **UART:** Universal Asynchronous Receiver/Transmitter

RELEVANT COURSES

- Verilog HDL, Digital IC design, Biomedical IC design, Advanced digital IC design, Machine Learning Hardware Design
- RTL Design and Verification, VLSI Design and Verification, FPGA-Based Digital System Design, Digital System Architecture
- CMOS digital VLSI design, Analog VLSI design, Semiconductor Device Physics.
- Design for testability, yield, and reliability, Real-time signal processing, and Semiconductor Device Modelling.

PROJECTS

Ph.D Projects:

1. CORDIC-Based Efficient VLSI Architecture for the EMD Algorithm with FPGA Demonstration

- Designed and developed an **efficient VLSI architecture for Empirical Mode Decomposition (EMD)** using the **CORDIC algorithm** for hardware-efficient computation.
- Developed the proposed architecture at the **RTL level** and evaluated its suitability for high-performance digital signal processing applications.
- Implemented and validated the architecture on an **FPGA platform**, demonstrating functional correctness and hardware feasibility.
- Performed **RTL simulation, synthesis, verification, and FPGA implementation** to evaluate design performance.
- Analyzed key hardware metrics, including **area, power, timing, and computational efficiency,** and explored architectural optimization techniques.

2. CORDIC-Based Low-Complexity Hardware Architecture for EMD-FastICA Algorithm for sEMG Signal Separation

- Designed a **low-complexity hardware architecture** for **sEMG signal separation** based on the integration of **Empirical Mode Decomposition (EMD)** and **Fast Independent Component Analysis (FastICA)**.
- Developed **CORDIC-based computational architectures** to reduce hardware complexity and efficiently implement computationally intensive mathematical operations.
- Designed and optimized the proposed architecture at the **RTL level** for FPGA-based implementation.
- Performed **RTL simulation, functional verification, synthesis, and FPGA implementation** to validate the proposed architecture.
- Evaluated the architecture in terms of **hardware resource utilization, timing performance, computational complexity, and signal-separation performance**.
- Targeted the proposed architecture toward **real-time biomedical signal processing applications**, particularly sEMG-based signal analysis.

3. CORDIC-Based Low-Complexity Hardware Design Methodology for K-Means Clustering Algorithm for sEMG Movement Classification with FPGA Implementation

- Designed a **CORDIC-based, low-complexity hardware architecture** for the **K-Means clustering algorithm** targeting real-time **sEMG movement classification**.
- Developed an efficient **RTL architecture** incorporating CORDIC-based computation to implement the required mathematical operations with reduced hardware complexity.
- Optimized the architecture for **low hardware resource utilization, computational efficiency, and real-time operation**.
- Implemented and validated the proposed architecture on an **FPGA platform** for sEMG-based movement classification.
- Performed **RTL simulation, functional verification, synthesis, and FPGA hardware testing** to validate the proposed design.
- Evaluated the architecture based on **FPGA resource utilization, timing performance, computational complexity, and classification accuracy**.

4. Low-Complexity, Low-Power Hardware-Software Co-Design Architecture for sEMG Signal Processing and Movement Classification in Prosthetic Applications

- Designed and demonstrated an **FPGA-based hardware-software co-design architecture** for real-time **sEMG signal processing and movement classification** targeting prosthetic applications.
- Developed the **Programmable Logic (PL) processing pipeline** with custom hardware IPs for computationally intensive sEMG processing, including **EMD-FastICA-based signal separation** and **CORDIC-based K-Means clustering**.
- Developed the **EMD/EMD-FastICA IP as a custom hardware accelerator** for the PL side, enabling efficient sEMG signal preprocessing and signal separation.
- Designed and integrated a **K-Means clustering IP** in the PL side for hardware-accelerated movement classification.
- Implemented **feature extraction and system-level processing on the Processing System (PS) side**, integrating the PS and PL through an FPGA-based hardware-software co-design framework.
- Demonstrated the complete architecture on **FPGA hardware using real-time/acquired sEMG signals**, evaluating classification performance, hardware resource utilization, processing latency, and power/complexity considerations.

Key Hardware IPs Developed as Part of the PhD Research:

- **CORDIC-Based EMD VLSI Architecture** - Efficient hardware architecture for Empirical Mode Decomposition.
- **CORDIC-Based EMD-FastICA IP** - Low-complexity hardware accelerator for sEMG signal separation.
- **CORDIC-Based K-Means IP** - Hardware accelerator for sEMG movement classification.
- **sEMG Feature Extraction & Classification Framework** - PS-side software processing integrated with PL-based hardware accelerators.

MTech Title:

COCHLEA-INSPIRED RF CHANNELIZED PRESELECTOR FILTER WITH RECEIVER FRONT END FOR UWB

AND ZIGBEE APPLICATIONS.

A planar manifold channelizer design is where the channelizer circuit is obtained from an electrical, mechanical analogy

of a

cochlear model. The electrical circuit's critical characteristics are identified, including channel filter input impedance and

manifold structure. They are applied to various technologies, producing several cochlea-like channelizers ranging from 2 GHz

MHz to 10.5 GHz.

Title:

Develop a Verilog-based UART (Universal Asynchronous Receiver and Transmitter) and demonstrate how it

works on FPGA Implementation.

The UART transmitter controls transmission by fetching a data word in parallel format and directing the UART to transmit it

in a serial format. Likewise, the Receiver must detect transmission, receive the data in serial format, strip off the start and

stop bits, and store the data word in a parallel format. Since the UART is asynchronous in working, the Receiver does not

know when the data will come, so the Receiver generates a local clock to synchronize with the transmitter whenever the start

bit is received.

Title:

Develop a Verilog-based Hamming code and demonstrate how it works on an FPGA implementation.

A commonly known linear Block Code is the Hamming code. Hamming codes can detect and correct a single-bit error in a

data block. The presence and location of a single parity bit error can be determined by analyzing parities of combinations of

received bits to produce a table of parities, each corresponding to a particular bit error combination. This table of errors is

known as the error syndrome. If all parity is correct according to this pattern, it can be concluded that there is not a single bit

error in the Messages.

Title:

RF LOW-NOISE AMPLIFIER DESIGN FOR 2.4G Hz WIRELESS ZigBee TRANSCIEVER APPLICATION.

Description: characterization of a low noise amplifier (LNA) operating at 2.4 GHz and implemented in a 180nm CMOS IBM

technology. Inductive source degeneration topology has been used because of its good trade-off between input matching and

noise figures. Simulation results show a gain of 17.96 dB and a Noise Figure of 2.58 dB for a power consumption of 11.7 mW.

All the design steps have been presented: gain optimization and bias point analysis, input and output matching, insertion of a

cascade stage, stability, noise, Figure analysis, and layout design.

EXTRACURRICULAR

- Awarded third prize in the Idea presentation on MEMS Devices conducted in the GIAN workshop on BIOMEMS Design.
- Participated in "INTER NIT KHO-KHO 2019" held at NIT Rourkela.
- Participated in athletic events, Inter IIT sports meet 2019 at IIT Kharagpur.
- Participated in athletic events, the Inter IIT sports meet 2022 at IIT Delhi.

PROFESSIONAL SUMMARY

PhD Research Scholar specializing in **RTL Design, FPGA-Based Digital System Design, and VLSI Architecture**, with hands-on experience in developing low-complexity and low-power hardware architectures for **signal processing and machine-learning applications**. Proficient in **Verilog/SystemVerilog, RTL design and verification, FPGA implementation, synthesis, and hardware-software co-design**. Experienced in developing and integrating custom **FPGA IP cores and hardware accelerators** for EMD, FastICA, CORDIC, and K-Means algorithms, with demonstrated implementation on FPGA platforms for real-time sEMG signal processing and prosthetic applications. Strong background in **digital IC design, ML hardware design, biomedical signal processing, and hardware optimization**, with experience leading technical projects and mentoring student teams.

PUBLICATIONS

- Cheduluri, Ganesh, et al. "Low complexity hardware architecture design methodology for cubic spline interpolation technique for assistive technologies." *2022 20th IEEE Interregional NEWCAS Conference (NEWCAS)*. IEEE, 2022.
- Patent1(Granted): LOW-Complexity Hardware System and a Method for Empirical Mode Decomposition of Signals, Patent No. : 599211
- Patent2(under examination): System and Method for Real-Time Clustering Signal Feature Data, No. **202641044553** FER reported.
- Patent 3(under examination): Hardware signal-processing system and method for real-time blind source separation, Indian Patent Application no. **202641103590**
- Patent 4(filed): Low-Complexity, Low-Power Hardware Software Co-Design Architecture of Surface EMG Signal Processing System for Movement Classification in Prosthetic Applications.
- Bhardwaj, Swati, Diptasri Ghosh, Debeshi Dutta, **Ganesh Cheduluri**, Vidhumauli Hansigida, Appa Rao Nali, and Amit Acharyya. "Low complex cordic-based hand movement recognition design methodology for rehabilitation and prosthetic applications." In *2023, the 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pp. 1-5. IEEE, 2023.